Discussion. Parametric g-formula: Outcome modeling

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causal3900.github.io

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Statistical modeling

Under exchangeability,

$$E\left(Y^a \mid \vec{L} = \vec{\ell}\right) = E\left(Y^a \mid A = a, \vec{L} = \vec{\ell}\right)$$

Under consistency,

$$E\left(Y^a \mid A = a, \vec{L} = \vec{\ell}\right) = E\left(Y \mid A = a, \vec{L} = \vec{\ell}\right)$$

To estimate, we have been taking the subgroup mean

$$\hat{E}(Y \mid A = a, \vec{L} = \vec{\ell}) = \frac{1}{n_{a,\vec{\ell}}} \sum_{i:A_i=a,\vec{L}_i=\vec{\ell}} Y_i$$

What happens when a subgroup is empty, we have continuous treatment variables or continuous covariates?

We need a model!

Outcome modeling: Parametric and Non-Parametric g-formula:

1. Learn a model to predict Y given $\{A, \vec{L}\}$
2. For each i, predict
   - $\{A = 1, \vec{L} = \vec{\ell}_i\}$, the conditional average outcome under treatment
   - $\{A = 0, \vec{L} = \vec{\ell}_i\}$, the conditional average outcome under control
3. Take the difference for each unit
4. Average over the units

Estimating a Model - Example

- Question: Does studying more that 7 hours increase the probability of passing an exam?
- Define the variables:
  - Y: 1 if pass, 0 if fail
  - A: number of hours studying
  - L_1: grade in Intro to stat
  - L_2: hours slept the night before

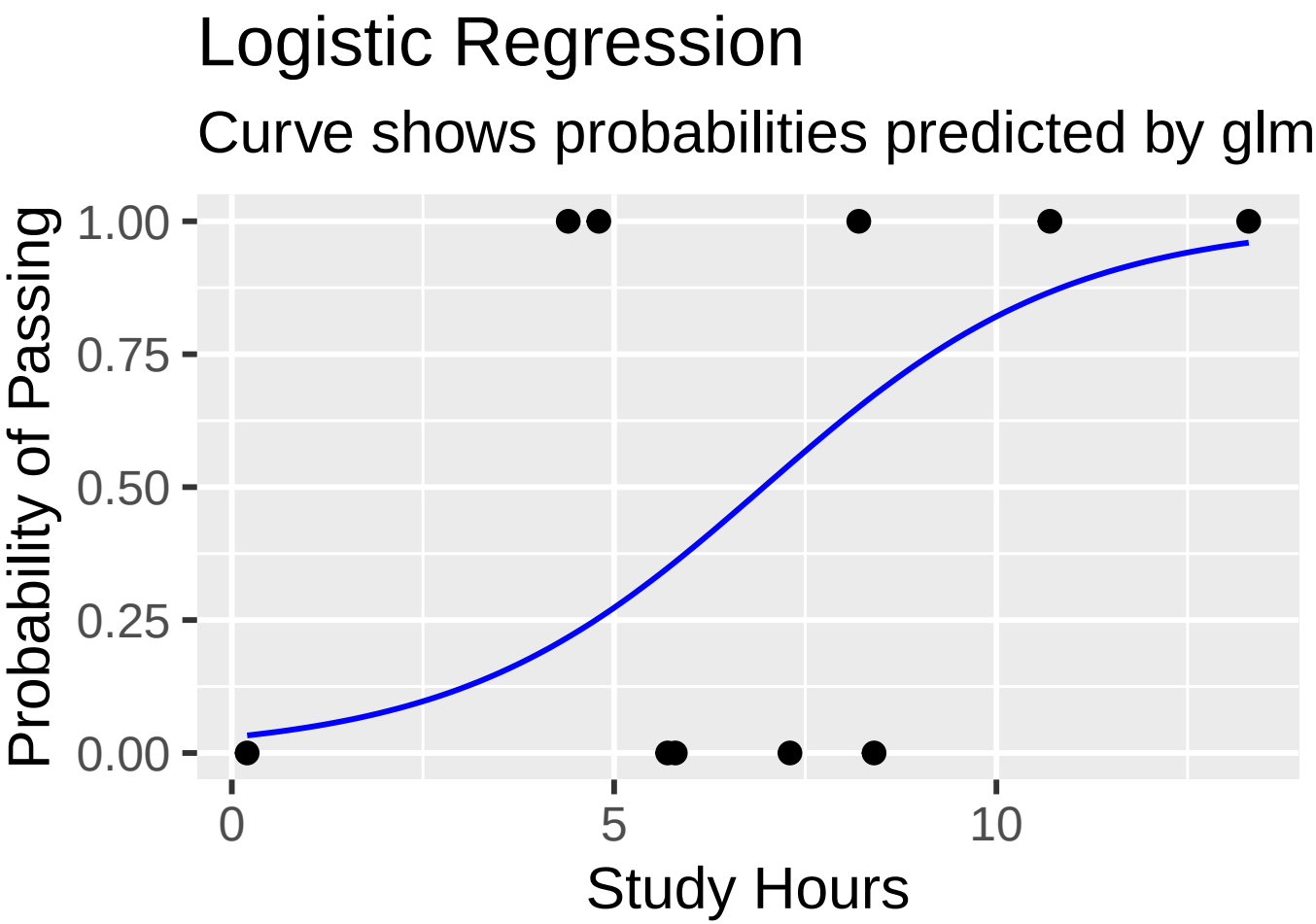
Example

Y	A	intro_grade	sleep_hours
0	5.7	3.81	7.7
0	8.4	1.70	6.2
0	0.2	2.30	3.8
0	7.3	3.93	6.7
1	13.3	3.90	6.4
1	4.8	1.45	7.8
1	4.4	1.83	9.5
1	10.7	1.89	9.8
0	5.8	3.85	7.8
1	8.2	3.42	5.4

Example

Learn a logistic regression model with `glm`:

```
model <- glm(Y ~ A + intro_grade + sleep_hours,
             data = exam_data,
             family = binomial(link = "logit"))
```



Example

We will now estimate the potential probability of passing the exam under treatment and no treatment for each individual. For this we converted the treatment to a binary variable where $A_i = 1$ if individual i studies more than 7 hours

Y	A	intro_grade	sleep_hours	P_yhat1	P_yhat0
0	0	3.81	7.7	0.687	0.258
0	1	1.70	6.2	0.732	0.302
0	0	2.30	3.8	0.252	0.051
0	1	3.93	6.7	0.499	0.136
1	1	3.90	6.4	0.450	0.115
1	0	1.45	7.8	0.909	0.614
1	0	1.83	9.5	0.964	0.810
1	1	1.89	9.8	0.970	0.836
0	0	3.85	7.8	0.697	0.267
1	1	3.42	5.4	0.349	0.078

Example

Finally to estimate the average treatment effect we will take the difference for each unit and average over them

```
exam_data %>%
  mutate(ATE = P_yhat1 - P_yhat0) %>%
  select(P_yhat1,P_yhat0,ATE) %>%
  summarise_all(.funs = mean) %>% kbl(digits = 3)
```

P_yhat1	P_yhat0	ATE
0.651	0.347	0.304

Bias Variance trade-off

Research question

To what degree does finishing college increase the probability of having a spouse who finished college?

Data. National Longitudinal Survey of Youth 1997

- Probability sample of U.S. non-institutional civilian youth age 12–16 on Dec 31 1996
- Surveyed annually 1997–2011, then biennially
- $n = 7{,}771$

G-formula: Data example

Estimate a model based on the true data

```
# A tibble: 10 x 4
   a          y     sex    race
   <chr>      <lgl> <chr>  <fct>
 1 college    FALSE Female Non-Hispanic Non-Black
 2 college    FALSE Female Non-Hispanic Non-Black
 3 college    TRUE  Male   Non-Hispanic Non-Black
 4 college    TRUE  Male   Non-Hispanic Non-Black
 5 no_college FALSE Male   Hispanic
 6 no_college FALSE Female Hispanic
 7 no_college TRUE  Male   Hispanic
 8 no_college FALSE Female Hispanic
 9 no_college FALSE Male   Hispanic
10 no_college FALSE Female Hispanic
```

Predict values - control

Predict the counterfactuals when everybody is in the control group

```
# A tibble: 10 x 3
   a           sex    race
   <chr>       <chr>  <fct>
 1 no_college Female Non-Hispanic Non-Black
 2 no_college Female Non-Hispanic Non-Black
 3 no_college Male   Non-Hispanic Non-Black
 4 no_college Male   Non-Hispanic Non-Black
 5 no_college Male   Hispanic
 6 no_college Female Hispanic
 7 no_college Male   Hispanic
 8 no_college Female Hispanic
 9 no_college Male   Hispanic
10 no_college Female Hispanic
```

Predict values - treatment

Predict the counterfactuals when everybody is in the treatment group

```
# A tibble: 10 x 3
   a       sex    race
   <chr>   <chr>  <fct>
 1 college Female Non-Hispanic Non-Black
 2 college Female Non-Hispanic Non-Black
 3 college Male   Non-Hispanic Non-Black
 4 college Male   Non-Hispanic Non-Black
 5 college Male   Hispanic
 6 college Female Hispanic
 7 college Male   Hispanic
 8 college Female Hispanic
 9 college Male   Hispanic
10 college Female Hispanic
```

1. Learn a model to predict Y given $\{A, \vec{L}\}$

- Parametric model:

```
fit_glm <- glm(y ~ a + sex + race + mom_educ +
                   dad_educ + log_parent_income +
                   log_parent_wealth +
                   test_percentile,
               data = d, family = binomial)
```

- Non-Parametric model:

```
fit_rf <- randomForest(
  y ~ a + sex + race + mom_educ + dad_educ +
    log_parent_income + log_parent_wealth +
    test_percentile,
  data = d %>% mutate(a = as.factor(a),
                      y= as.factor(y)),
  ntree = 500)
```

2. Predict conditional average potential outcomes

- Parametric model:

```
average_outcomes_glm <- d %>%
  mutate(yhat1 = predict(fit_glm,
                         newdata = d %>%
                           mutate(a = "college"),
                         type = "response"),
         yhat0 = predict(fit_glm,
                         newdata = d %>%
                           mutate(a = "no_college"),
                         type = "response"))
```

2. Predict conditional average potential outcomes

Notice how the `predict` function adapts to the model specified, pay attention to its argument and output - Non-Parametric model:

```
average_outcomes_rf <- d %>%
  mutate(yhat1 = predict(fit_rf,
                         newdata = d %>%
                           mutate(a = factor("college",
                         levels =c("college","no_college"))),
                         type="prob")[,"TRUE"],
         yhat0 = predict(fit_rf,
                         newdata = d %>%
                           mutate(a = factor("no_college",
                         levels = c("college","no_college"))),
                         type="prob")[,"TRUE"])
```

3. Difference to estimate conditional average effects

- Parametric model:

```
average_effects_glm <-
  average_outcomes_glm %>%
  mutate(effect = yhat1 - yhat0)
```

- Non-Parametric model:

```
average_effects_rf <-
  average_outcomes_rf %>%
  mutate(effect = yhat1 - yhat0)
```

4. Average over units

- Parametric model:

```
average_effects_glm %>% select(yhat1, yhat0, effect) %>%
  summarize_all(.funs = mean)
```

```
# A tibble: 1 x 3
  yhat1 yhat0 effect
  <dbl> <dbl>  <dbl>
1 0.381 0.163  0.218
```

- Non-Parametric model:

```
average_effects_rf %>% select(yhat1, yhat0, effect) %>%
  summarize_all(.funs = mean)
```

```
# A tibble: 1 x 3
  yhat1 yhat0 effect
  <dbl> <dbl>  <dbl>
1 0.392 0.123  0.269
```

G-formula: Parametric vs. Non-Parametric

- Parametric models have lower variance but probably high bias
- Non-parametric models have lower bias but higher variance
- Non-parametric methods usually require more observations
- Note that each time you run `randomForest` the results might slightly change

Recap. Parametric g-formula: Outcome modeling

1. Learn a model to predict Y given $\{A, \vec{L}\}$
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